

Example

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1 Quarto Example

This is an example of a Quarto document, you will need to install Quarto from <https://quarto.org/>. This package comes with several common R packages, it is also recommended to install:

```
install.packages(c('gt', # general table formatting
  'gtsummary', # for univariate and regression tables
  'here', # where you are working (better when working with git)
  'knitr', # r package for working with documents
  'quarto', # R package for working with quarto
  'rmarkdown', # creating documents
  'tidymodels', # modern interface for fitting models
  'tidyverse', # modern R language
  'tinytex', # needed for pdf documents
  'devtools' # package creation development tools
))

remotes::install_github("Pakillo/grateful") # for package citation
```

To create pdf documents it is also recommended to run

```
tinytex::install_tinytex()
```

Table 1: first 5 records and 5 columns of the data

arl_number	head_shape	orientation	nom_velocity_m_sec	init_strike_velocity_m_s
11	Hemis	CW	40	39
12	Hemis	CW	40	40
13	Hemis	CW	40	39
14	Hemis	CW	40	39
15	Hemis	CW	40	40

2 Table One

Table 2: An example of a Table 1

Characteristic	Flatter, N = 13 ¹	Hemis, N = 40 ¹	p-value	Overall ¹
orientation			0.067 ²	
CW	4 (31%)	24 (60%)		28 (53%)
Spine	9 (69%)	16 (40%)		25 (47%)
nom_velocity_m_sec			0.002 ³	
40	13 (100%)	21 (53%)		34 (64%)
70	0 (0%)	19 (48%)		19 (36%)
init_strike_velocity_m_s			0.13 ⁴	
Mean (SD)	41 (1)	55 (16)		51 (15)
Median (IQR)	41 (40, 42)	42 (40, 71)		42 (40, 70)
Median (Range)	41 (38 - 43)	42 (39 - 75)		42 (38 - 75)
depth_mm			0.73 ⁴	
Mean (SD)	31 (11)	32 (12)		32 (12)
Median (IQR)	29 (24, 37)	30 (21, 39)		30 (21, 39)

Median (Range)	29 (17 - 53)	30 (18 - 59)		30 (17 - 59)
max_depth_mm			0.94 ⁵	
Mean (SD)	34 (12)	33 (12)		33 (12)
Median (IQR)	31 (25, 43)	31 (22, 40)		31 (23, 41)
Median (Range)	31 (19 - 57)	31 (19 - 59)		31 (19 - 59)
duration_ms			0.73 ⁴	
Mean (SD)	1.00 (0.40)	0.96 (0.41)		0.97 (0.40)
Median (IQR)	0.90 (0.73, 1.23)	0.85 (0.62, 1.21)		0.87 (0.62, 1.23)
Median (Range)	0.90 (0.52 - 1.81)	0.85 (0.51 - 2.00)		0.87 (0.51 - 2.00)
transf_energy_j			0.95 ⁵	
Mean (SD)	124 (41)	135 (84)		130 (68)
Median (IQR)	128 (93, 160)	115 (82, 180)		126 (89, 163)
Median (Range)	128 (67 - 190)	115 (29 - 297)		126 (29 - 297)
Unknown	1	24		25
total_impulse_ns			0.45 ⁵	
Mean (SD)	3.40 (1.27)	2.96 (1.71)		3.15 (1.53)
Median (IQR)	3.45 (2.43, 4.47)	2.48 (1.49, 4.72)		2.96 (1.80, 4.57)
Median (Range)	3.45 (1.74 - 5.72)	2.48 (0.55 - 5.36)		2.96 (0.55 - 5.72)
Unknown	1	24		25
force_kn			0.30 ⁵	
Mean (SD)	7.16 (1.48)	7.20 (3.82)		7.19 (3.00)
Median (IQR)	6.82 (5.87, 8.21)	5.22 (4.49, 9.84)		6.19 (5.01, 8.74)
Median (Range)	6.82 (5.37 - 10.26)	5.22 (2.83 - 15.47)		6.19 (2.83 - 15.47)
Unknown	1	24		25
ang_pp_chest_wall_deg			0.095 ⁴	
Mean (SD)	-13 (9)	-9 (6)		-11 (7)
Median (IQR)	-17 (-20, -2)	-10 (-14, -4)		-11 (-16, -4)
Median (Range)	-17 (-24 - -2)	-10 (-19 - 1)		-11 (-24 - 1)
Unknown	0	12		12
ang_pp_spine_degree			0.077 ⁴	
Mean (SD)	6.5 (6.8)	10.5 (6.8)		9.2 (7.0)
Median (IQR)	3.0 (1.0, 14.0)	11.0 (5.0, 14.8)		8.0 (3.0, 14.0)
Median (Range)	3.0 (0.0 - 18.0)	11.0 (-2.0 - 25.0)		8.0 (-2.0 - 25.0)
Unknown	0	12		12
aa_traj_chest_wall_deg			0.095 ⁴	
Mean (SD)	77 (9)	81 (6)		79 (7)
Median (IQR)	73 (70, 88)	80 (77, 86)		79 (74, 86)
Median (Range)	73 (66 - 88)	80 (71 - 91)		79 (66 - 91)
Unknown	0	12		12
aa_traj_spine_degree			0.077 ⁴	
Mean (SD)	96.5 (6.8)	100.5 (6.8)		99.2 (7.0)
Median (IQR)	93.0 (91.0, 104.0)	101.0 (95.0, 104.8)		98.0 (93.0, 104.0)
Median (Range)	93.0 (90.0 - 108.0)	101.0 (88.0 - 115.0)		98.0 (88.0 - 115.0)
Unknown	0	12		12

¹n (%)

²Pearson's Chi-squared test

³Fisher's exact test

⁴Wilcoxon rank sum test

⁵Wilcoxon rank sum exact test

3 Cross Reference

You can cross-reference tables, listings, figures and page numbers

You can use the `labels` you created to reference the tables like Table 2 or items in the `bib.bib` file such as Al-Helali et al. (2021).

You can even reference later items such as Figure 2 or sections like Section 5 or even the page number that something happens on like page 7.

4 Pairwise t-tests

Suppose we wanted to visualize all pairwise t-tests (we will only do first four but you get the idea). There are other ways to do this but this is a good enough example of repeated analysis.

4.1 between init_strike_velocity_m_s and depth_mm

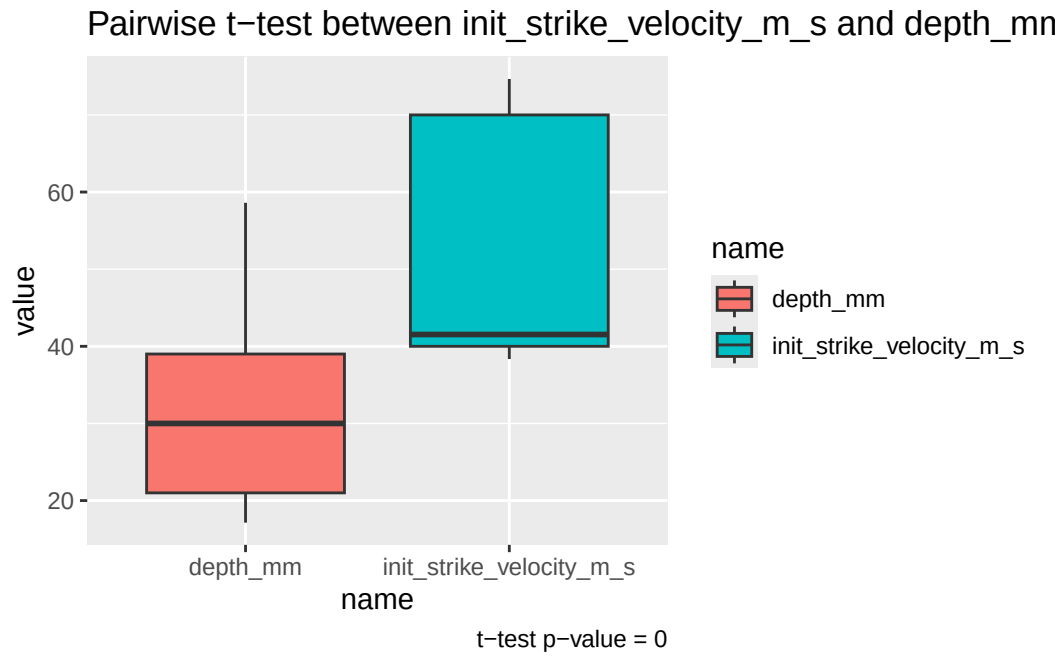


Figure 1: Pairwise t-test between init_strike_velocity_m_s and depth_mm

4.2 between init_strike_velocity_m_s and max_depth_mm

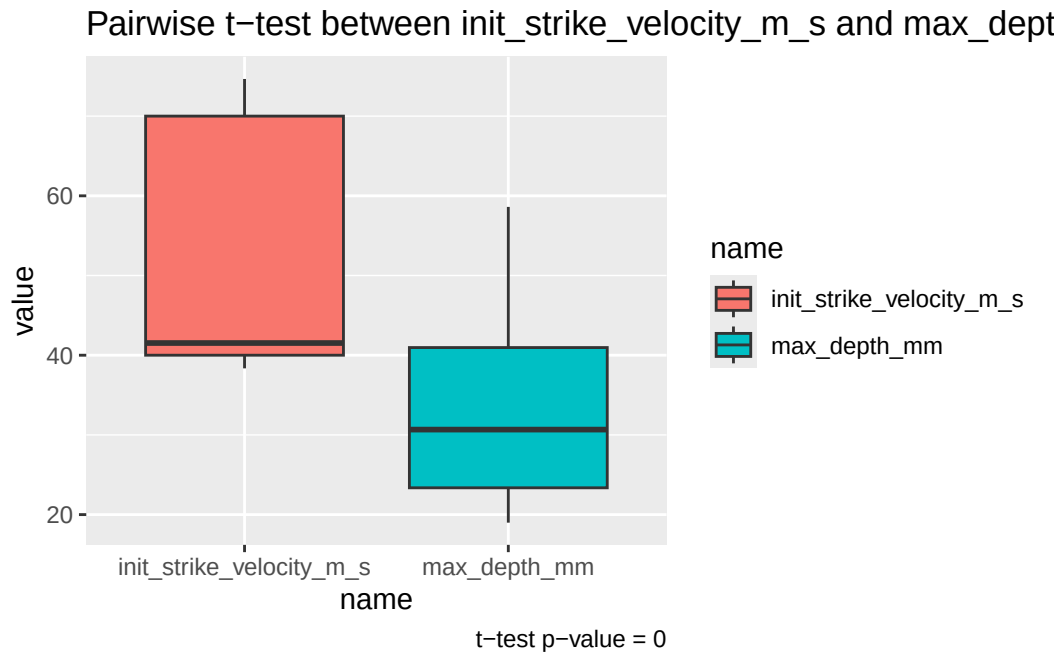


Figure 2: Pairwise t-test between init_strike_velocity_m_s and max_depth_mm

4.3 between init_strike_velocity_m_s and duration_ms

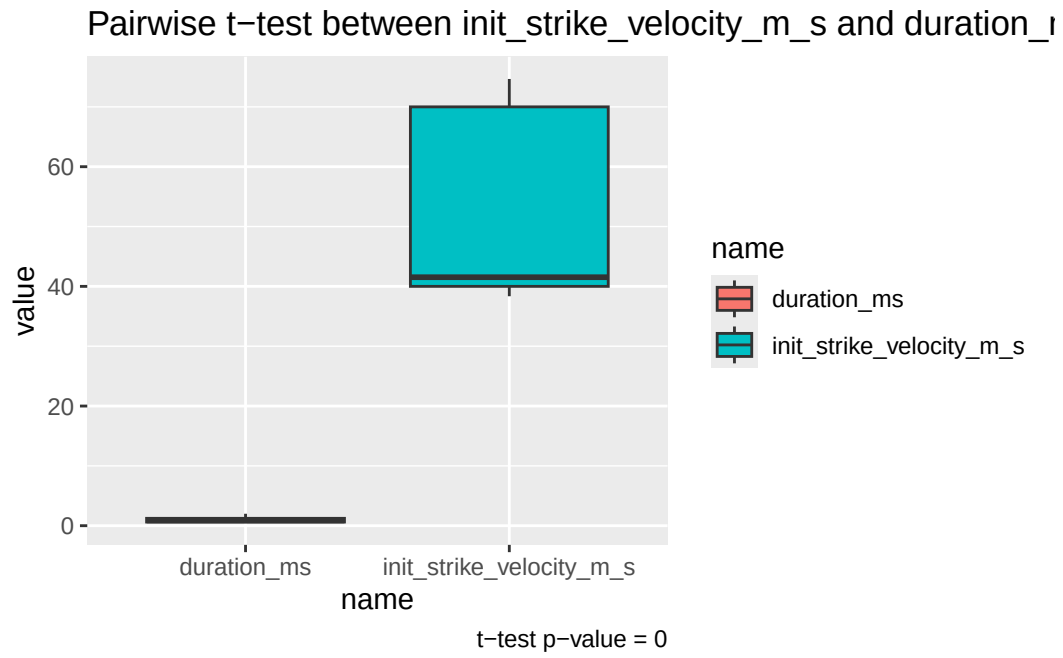


Figure 3: Pairwise t-test between init_strike_velocity_m_s and duration_ms

4.4 between init_strike_velocity_m_s and transf_energy_j

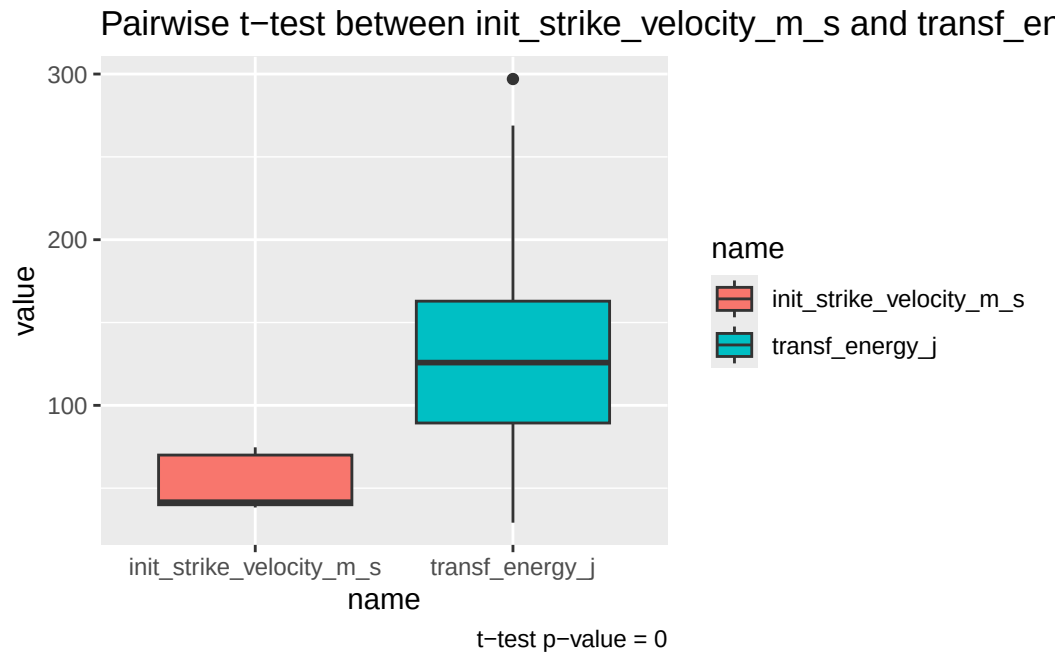


Figure 4: Pairwise t-test between init_strike_velocity_m_s and transf_energy_j

5 Software

This work was completed using R version 4.4.0 (R Core Team 2024) with the following R packages: bookdown v. 0.39 (Xie 2016, 2024a), datasci v. 0.1.0 (Armstrong 2024), grateful v. 0.2.9 (Rodriguez-Sanchez and Jackson 2023), gt v. 0.10.1 (Iannone et al. 2024), gtsummary v. 1.7.2 (Sjoberg et al. 2021), here v. 1.0.1 (Müller 2020), knitr v. 1.47 (Xie 2014, 2015, 2024b), rmarkdown v. 2.27 (Xie, Allaire, and Golemund 2018; Xie, Dervieux, and Riederer 2020; Allaire et al. 2024), tidyverse v. 2.0.0 (Wickham et al. 2019).

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